

Interrelation between the Vegetation Index and the Climatic Parameters and Structural Characteristics of Vegetation Cover

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Abstract—The interrelations between the normalized difference vegetation index and the total incoming radiation; total evaporation; mean surface-air temperature over the vegetation period; and the structural characteristics of vegetation cover, such as its overground phytomass and overground net primary production, are analyzed on the basis of experimental and satellite data. The analysis was made with the use of the data obtained for plain regions of European Russia and western Siberia. It is shown that, for these regions, the latitudinal distributions of the values of the vegetation index and the structural vegetation-cover characteristics under consideration do not coincide. The relationships between the structural characteristics of vegetation cover (phytomass, production) and the vegetation index have the form of functions with saturation. The relationships between the vegetation index and the climatic parameters under consideration have a bell-shaped form. The analysis has shown that the normalized difference vegetation index is not an indicator for the values of the overground net primary production of vegetation cover and its overground phytomass within the areas under consideration.

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INTRODUCTION

The vegetation cover of terrestrial ecosystems plays a significant part in the planet's climate system. Changes in the cover's properties cause variations in the radiation characteristics of the land surface, its roughness, and in the dynamics of exchange of heat and moisture in the atmosphere–underlying surface system. The energy exchange between the land surface and the atmosphere, as well as biogeochemical processes, changes on both global and regional scales [1–3].

With the advance of the system of satellite monitoring of the climate system's radiation fields, the remote methods of analyzing and mapping terrestrial vegetation on local, regional, and global levels were developed. The direct measurements of structural vegetation-cover characteristics, such as phytomass, net primary production, and leaf area index, are practically impossible for large areas. These quantities are usually measured at experimental sites, and the results obtained are used for comparatively similar (in climate and vegetation) regions. At present, an analysis of the characteristics of the underlying surface with the aid of remote methods based on its reflectivity allows one to obtain data on rather extensive areas. However, the reflectivity of the underlying surface is

affected by a multitude of factors: the phenological state of vegetation cover, the dominant type of vegetation, the type of phytocenosis, the degree of its damage, observation conditions (the Sun's altitude and observation angle), illumination (portions of direct and scattered radiation and atmospheric transparency), weather conditions (cloudiness, atmospheric humidity, and air temperature), the type and moisture of soil, etc. The accuracy of data obtained from satellite monitoring of vegetation cover depends on a numerous factors of technical character as well, including the correctness of considering for the Sun's zenith angle, observation geometry, the drift of instrumentation calibration, and the drift of the satellite's orbit.

Vegetation absorbs solar radiation mainly in the visible (red, 0.58–0.68 μm) region of the solar spectrum and reflects and scatters it in the near infrared (0.73–1.1 μm) region. The electromagnetic optical spectrum of the vegetation canopy changes with the type of vegetation and its growth phase under different environmental conditions. This property of vegetation cover is used in remote methods for analysis of the development of vegetation and its state. The results of measurements with advanced very high resolution radiometers mounted aboard the NOAA satellites are

used most frequently. The data on the outgoing radiation obtained with radiometers in both the visible and near infrared regions of the electromagnetic spectrum are used to describe the structural characteristics of vegetation cover. To this end, a number of different algebraic combinations of the results of satellite measurements within the indicated spectral regions were proposed and called vegetation indices [4, 5].

In this study, the interrelations between the most universal index of vegetation activity of land plants—the normalized difference vegetation index—on the one hand and a number of climatic parameters and phytometric characteristics of vegetation cover on the other hand are analyzed on the basis of experimental and satellite data.

NORMALIZED DIFFERENCE VEGETATION INDEX AND STRUCTURAL CHARACTERISTICS OF VEGETATION COVER

The normalized difference vegetation index (NDVI) is defined as a normalized difference in brightness of the images of the underlying surface in the visible (red) (f_1) and near infrared (f_2) regions:

$$\text{NDVI} = \frac{f_2 - f_1}{f_1 + f_2}.$$

The values of this quantity may vary from -1 to 1 . For the underlying surface covered with green vegetation, f_2 is always greater than f_1 ; therefore, the NDVI values are positive. On the basis of satellite and ground-based measurements, it is found that, for plants in a normal state, the NDVI is close to 0.6 – 0.65 . Low NDVI values (on the order of 0.3 – 0.4) suggest a moisture deficiency or a plant disease; for deserts, the NDVI is lower than 0.02 [6, 7]. In a number of works [8–10], this index is treated as a set of environmental conditions that determine the rate of the production process of vegetation cover. The authors of these works believe that the NDVI is an integral index of vegetation activity of terrestrial vegetation.

After the NDVI values are calculated, a question arises in regards to how, on the basis of these values, to characterize the properties of the vegetation cover and, primarily, its structural characteristics such as its overground phytomass, overground net primary production (NPP), and leaf area index (LAI). The overground phytomass is determined by the amount of organic matter of overground vegetation per unit area. This quantity is calculated in the units of organic-matter dry weight and is measured, as a rule, in t/ha. The overground NPP is determined by the amount of organic matter grown over a certain period (usually over a year) per unit area in the overground vegetation

cover. The NPP is calculated in the units of organic-matter dry weight and is measured, as a rule, in t/ha per year. The methods of calculating the NPP are discussed in detail in [11]. The LAI shows how many times the one-sided leaf area exceeds the area of the minimum base of the vertical column within which these leaves are located.

Local statistical relationships between the NDVI and the production of some crops were established on the basis of ground-based measurements and satellite data. For example, in [12], the dependence of cotton yield (on a saline soil) on NDVI values is considered; and the ratios between the values of the overground phytomass of a number of grains and the NDVI are given in [13].

The production efficiency models (PEMs) allow one to calculate the absorbed (by vegetation cover) photosynthetically active radiation (APAR) on the basis of a satellite monitoring of the underlying surface and, thus, to relate satellite data to net primary production on both regional [14–16] and global [17–19] scales. The models of this type are based on the Kumar–Monteith algorithm [8], according to which a fraction of photosynthetically active radiation (FPAR) absorbed by vegetation cover is a function of the NDVI. On the basis of FPAR values and satellite data on incoming (to the underlying surface) photosynthetically active radiation, one can calculate the APAR. The productivity of vegetation cover is determined as a function of the APAR and the efficiency with which vegetation transforms the APAR to phytomass. This method is used for approximately calculating the spatial distribution of the NPP for vegetation of different types and for estimating the interseasonal dynamics of this quantity. In the PEMs, the structure of vegetation canopy and its phenology are determined as FPAR functions. Differences in model parametrizations and in the assumptions on the structure of vegetation and sensitivity to environmental factors result in both spatial and temporal variations in the estimates obtained for phytomass and net primary production [11, 20]. The coefficients of correlation between empirically measured and model values of overground phytomass and overground NPP vary from 0.5 to 0.9 [21].

The LAI is determined by the photosynthesizing part of plants. The LAI significantly affects variations in the spectral composition of reflected radiation and enters almost all models of a production process as one of the basic characteristics of vegetation cover. The NDVI dependences of the LAI were obtained from experimental and satellite data [9, 22]. On the basis of satellite data, the LAI was estimated (with consideration for the spatial distribution of leaves, their color, and soil effect) for different illumination conditions [23]. With the use of experimental data, it

was shown that the error in estimating the LAI from satellite data is insignificant (no more than 1%) for small values of the leaf index (up to $1 \text{ m}^2/\text{m}^2$). However, with an increase in the LAI, the error increases and becomes greater than 35% for LAI values from 3 to $5 \text{ m}^2/\text{m}^2$. In [24], it is noted that, if the LAI is higher than $6 \text{ m}^2/\text{m}^2$, there is no correlation between the LAI and the NDVI. In particular, the LAI characterizes the density of vegetation canopy and the degree of its projective cover. For example, in the middle latitudes of the Northern Hemisphere, the projective cover of vegetation reaches 100% if the LAI is $3 \text{ m}^2/\text{m}^2$ [25, 26]. In [27], it is noted that the NDVI has its maximum value at 40–60% of the projective cover of vegetation. In [28], it is concluded that, for a closed vegetation canopy, no changes in reflection occur in the nearest infrared spectral region. In [29], it is found that satellite data are informative only if the density of vegetation cover is less than 50%, i.e., if a vegetation–soil contrast exists. This condition is characteristic of arid and semiarid regions as well as regions with a developed vegetation cover at the beginning of the vegetation period. The NDVI data successfully used to study the vegetation cover of the arid and semiarid regions of the world, in particular, for mapping desertification and studying droughts [30, 31], confirm the conclusions made by the authors of [29].

RESEARCH METHODS AND ANALYSIS OF RESULTS

In this study, the mean NDVI values over the 1982–2000 vegetation periods were used on a spatial grid with $1^\circ \times 1^\circ$ resolution of geographic latitude and longitude [32]. The mean (over the vegetation periods) values of total evaporation are calculated on the basis of the data from [33]. The mean values of surface-air temperature over the periods under consideration are determined from the UEA/CRU database [34, 35]. The total solar radiation over the vegetation periods is calculated from the data given in [36].

Analysis of the interrelations between the NDVI and the climatic parameters and structural characteristics of vegetation cover is made for plain areas in European Russia and western Siberia. Mountainous regions are not considered because of additional optical effects (different mountain-slope illumination, shading effect, abrupt changes in temperature and radiation gradients, etc.) occurring in mountains and requiring special methods for their study. The areas under consideration are dissimilar in both climatic parameters and vegetation types. A large body of empirical data on net primary production and phytomass is available for these areas [37, 38].

The dependences of NDVI values on the climatic parameters and structural characteristics of vegetation

cover are obtained from comparison of data at each grid point for the areas under consideration.

Figure 1 gives the latitudinal distributions of the overground NPP and phytomass (from the data for the second half of the 20th century) and the latitudinal distributions of the NDVI for European Russia and western Siberia. The maximum NDVI values averaged over latitudes are almost the same (about 0.7) for the areas under consideration (Fig. 1a). For European Russia, these NDVI values are observed in the latitudinal belt 58° – 62° N. For western Siberia, the maximum NDVI values are characteristic of the narrower latitudinal belt 57° – 59° N. For European Russia, to the south of 58° N, the NDVI decreases approximately to 0.37 in the vicinity of 47° N and, then, increases up to 0.5 at 45° N. To the north of 62° N, the NDVI decreases gradually to 0.52 at about 69° N. For western Siberia, the general tendency of variation in NDVI values is similar to that for European Russia. To the south of 57° N, the NDVI decreases approximately to 0.59 in the latitudinal belt 54° – 53° N. To the south of 53° N, the NDVI decreases to 0.46 in the vicinity of 51° N. To the north of 59° N, the NDVI decreases gradually to 0.41 at about 70° N and, then, increases insignificantly up to 0.44 at 71° N. To the north of 71° N, the NDVI decreases to 0.27 in the vicinity of 76° N.

It follows from Fig. 1b that, for different regions of Russia, the empirical latitudinal dependences of the overground NPP differ significantly. For European Russia, the largest values of the NPP are observed in the latitudinal belt 54° – 59° N (its maximum of about 8.7 t/ha during the vegetation period is reached in the vicinity of 58° N). For western Siberia, the latitudinal belt with the largest NPP values is narrower (approximately 54° – 57° N) with its maximum of about 6 t/ha during the vegetation period in the vicinity of 56° – 57° N. An abrupt decrease of the NPP to the north and south of the latitudinal zone with its maximum values is characteristic of European Russia: 2.6 t/ha during the vegetation period in the vicinity of 50° N and 5.3 t/ha during the vegetation period in the vicinity of 61° N. In more southern latitudes of European Russia, an insignificant increase of the NPP up to 3.4 t/ha over the vegetation period is observed. To the north of 61° N, the NPP continues to decrease. This decrease is insignificant (to 4.5 over the vegetation period) in the latitudinal belt 61° – 65° N and is abrupt (to 1.7 over the vegetation period) in the latitudinal belt 68° – 69° N. An abrupt decrease of the NPP to the south of 55° N (to 3 t/ha over the vegetation period) in the vicinity of 51° N and a rather gradual decrease of the NPP to the north of 58° N (to 7.3 t/ha over the vegetation period) in the vicinity of 76° N are characteristic of western Siberia.

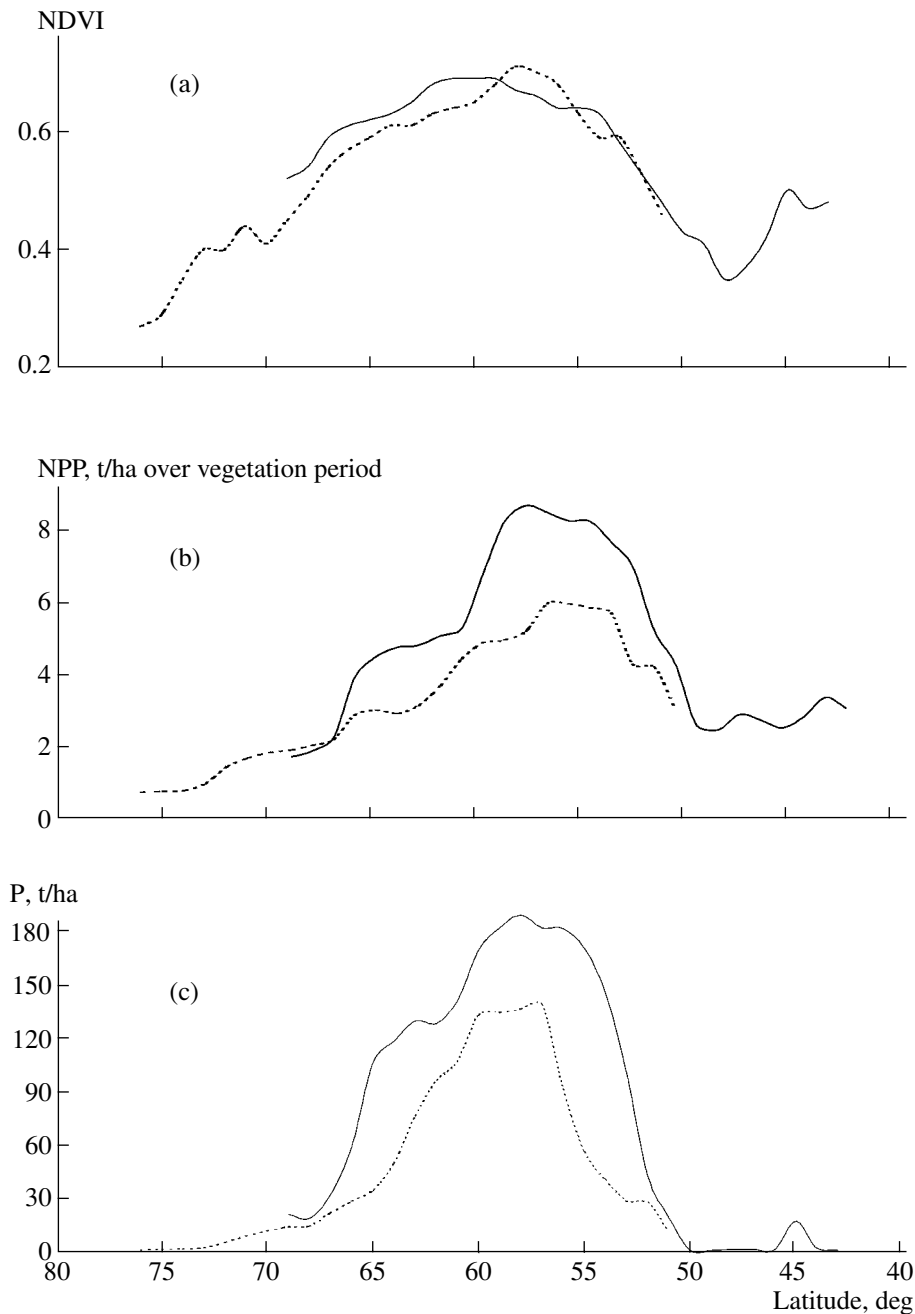


Fig. 1. Latitudinal distributions of (a) NDVI, (b) NPP, and (c) overground phytomass (P) for (solid curve) European Russia and (dashed curve) western Siberia.

It follows from Fig. 1c that the empirical latitudinal dependences of phytomass for the Russian regions under consideration differ substantially. For European Russia, the largest values of phytomass are observed in the latitudinal belt 55°–60° N with a maximum of about 190 t/ha in the vicinity of 58° N. For western Siberia, the latitudinal belt with the largest values of phytomass is narrower (approximately 57°–60° N) with a maximum of about 140 t/ha in the vicinity of 57° N. For European Russia, an abrupt decrease in

phytomass values to 2.1 t/ha is observed to the south of the latitudinal zone with maximum values in the vicinity of 50° N. Significant variations in phytomass values are observed to the south of 50° N only at 45° N, where the value increases to 18 t/ha owing to the areas covered with the North Caucasian foothill oak and hornbeam–oak forests. For European Russia, to the north of 60° N, the overground phytomass decreases approximately to 129–130 t/ha at 62°–63° N. To the north of 63° N up to subpolar latitudes, the phy-

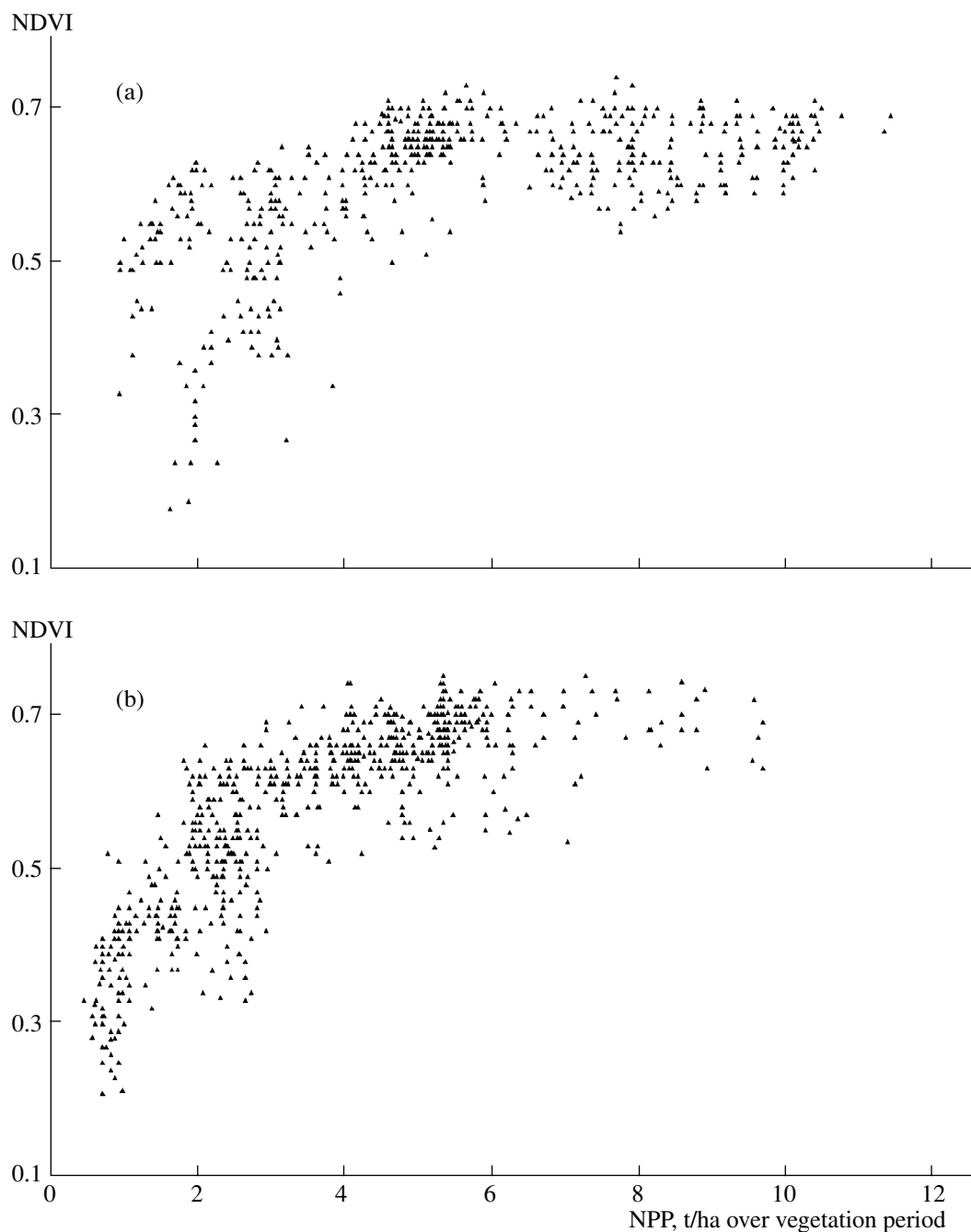


Fig. 2. Dependence of the NDVI on the NPP for (a) European Russia and (b) western Siberia.

tomass decreases to 22.1 t/ha. An abrupt decrease (to 12.3 t/ha) of the phytomass to the south of the latitudinal zone with a maximum value at 51° N is also characteristic of western Siberia. To the north of 60° N up to subpolar latitudes, a more gradual decrease in the phytomass to 2.1 t/ha is observed in the vicinity of 76° N.

It follows from the given graphs of the latitudinal distributions of the NPP, phytomass, and NDVI that variations in NDVI values are not similar to those in

NPP and phytomass values, i.e., to variations in structural characteristics of vegetation cover. Thus, one can assume that the NDVI is not significantly affected by the NPP and phytomass.

Figure 2 shows the dependences of the NDVI on the overground NPP for (a) European Russia and (b) western Siberia. A decrease in the range of NDVI variation with increasing NPP is characteristic of the presented sets of points. For European Russia, the NPP varies from 0.9 to 11.4 t/ha during the vegetation

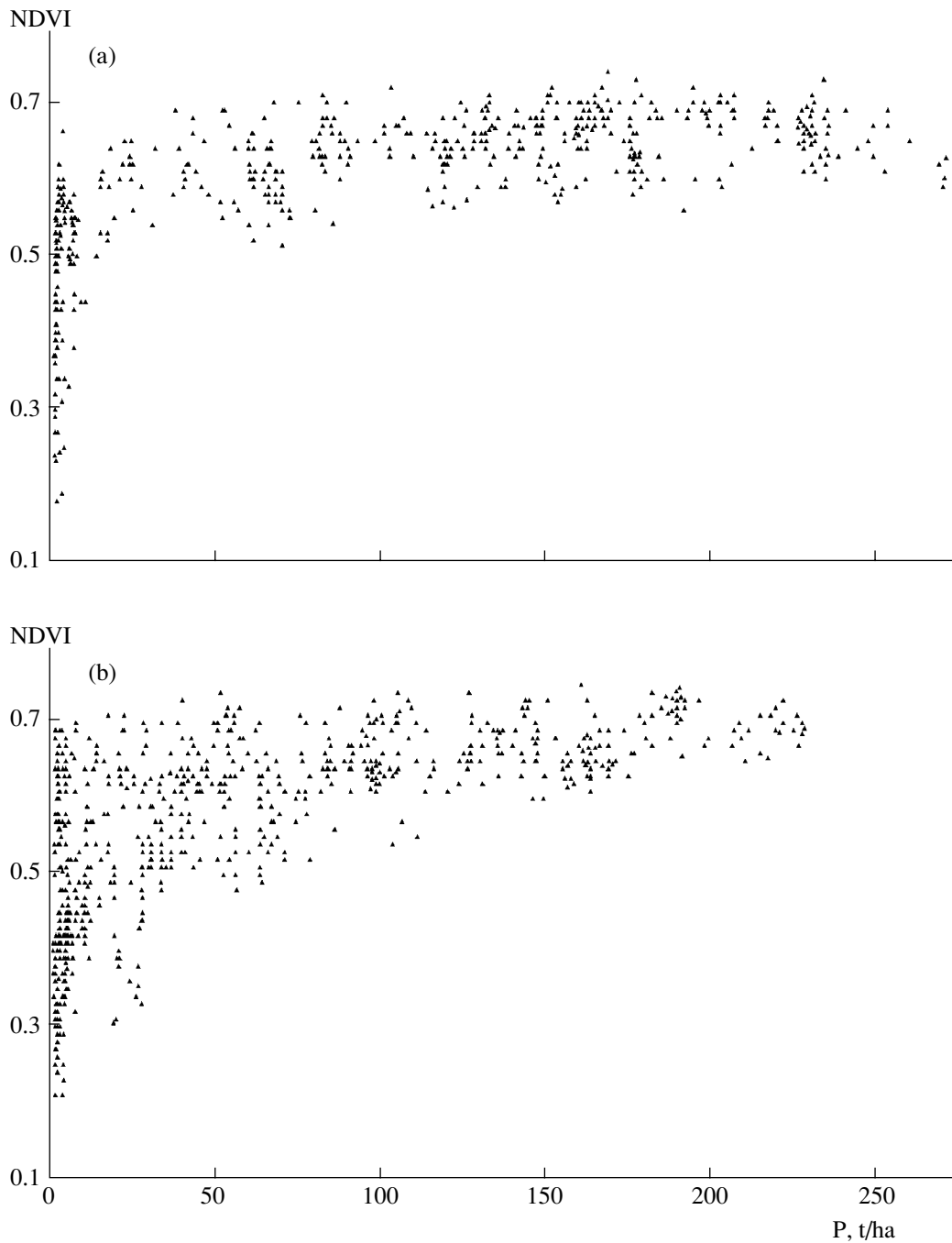


Fig. 3. Dependence of the NDVI on overground phytomass (P) for (a) European Russia and (b) western Siberia.

period, whereas the NDVI varies from 0.18 to 0.74. For western Siberia, the NPP varies from 0.5 to 9.7 t/ha during the vegetation period, while the NDVI varies from 0.21 to 0.75. A relatively uniform distribution of the points of the presented set over the whole range of NPP variation is characteristic of European Russia, while the distribution of most of the points of the set under consideration in the range of NPP varia-

tion from 0.5 to 6 t/ha over the vegetation period is characteristic of western Siberia.

Figure 3 gives the dependences of the NDVI on the overground phytomass for (a) European Russia and (b) western Siberia. The reserves of phytomass vary from 1.2 to 271.3 t/ha for European Russia and from 1.2 to 228.8 t/ha for western Siberia. For both European Russia and western Siberia, the largest range of

Table 1. Statistical characteristics of the NDVI for different ranges of variations in structural vegetation-cover characteristics: mean ($\bar{x}$), minimum ($x_{\min}$), and maximum ($x_{\max}$) values, rms deviation (σ_x), and variation coefficient (v_x , %)

Statistical characteristic	Range of variations in overground phytomass (t/ha)						Range of variations in overground production (t/ha over vegetation period)				
	0–10	10–30	30–70	70–112	112–180	180–250	0–2	2–4	4–6	6–8	8–10
European Russia											
$\bar{x}$	0.47	0.59	0.61	0.64	0.65	0.67	0.48	0.52	0.65	0.65	0.64
$x_{\min}$	0.18	0.44	0.52	0.51	0.56	0.56	0.18	0.24	0.50	0.54	0.56
$x_{\max}$	0.66	0.65	0.70	0.72	0.74	0.73	0.63	0.65	0.73	0.74	0.71
σ_x	0.10	0.05	0.04	0.05	0.04	0.03	0.12	0.08	0.04	0.04	0.04
v_x	21.3	8.8	6.7	7.0	5.9	5.0	25.1	16.1	6.5	6.4	6.3
Western Siberia											
$\bar{x}$	0.46	0.53	0.60	0.65	0.66	0.70	0.42	0.56	0.66	0.67	0.68
$x_{\min}$	0.21	0.31	0.48	0.52	0.60	0.65	0.21	0.33	0.52	0.54	0.63
$x_{\max}$	0.70	0.71	0.74	0.74	0.75	0.75	0.64	0.71	0.75	0.75	0.74
σ_x	0.12	0.10	0.06	0.05	0.03	0.02	0.09	0.08	0.04	0.06	0.04
v_x	25.1	19.0	10.4	7.6	4.8	3.4	21.0	14.3	6.7	8.7	5.1

NDVI variation is characteristic of phytomass from 1.2 to 10 t/ha. For this range of phytomass variation, the NDVI varies from 0.18 to 0.66 for European Russia and from 0.21 to 0.7 for western Siberia. The range of NDVI variation from 0.44 to 0.74 is characteristic of European Russia, with the phytomass varying from 10 to 271.3 t/ha. For western Siberia, when the phytomass varies from 10 to 30 t/ha, the NDVI varies within a sufficiently wide range from 0.31 to 0.71; when the phytomass varies from 30 to 112 t/ha, the range of NDVI variation becomes somewhat narrower (0.48–0.74). Under a further increase in phytomass values, the range of NDVI variation (0.6–0.75) becomes comparable to the range characteristic of analogous phytomass values for European Russia.

Table 1 presents NDVI statistical characteristics for different ranges of variation in the phytomass and primary production. It follows from the data given in Figs. 2 and 3 and Table 1 that, as the production and phytomass reserve increase during the vegetation period, the NDVI values rapidly reach the range of their small variation. This is due to the fact that the optical properties of vegetation cover depend on the ability of green leaves to absorb photosynthetically active radiation. Reflection from the vegetation canopy is related to the content of green pigments (chlorophylls) in the leaves of plants [39]. The dependence of the content of chlorophyll in the green parts of plants on the intensity of its radiation absorption is a

function with saturation. The NDVI is calculated on the basis of the reflecting characteristics of the underlying surface, and, therefore, the above dependences of this quantity on the NPP and phytomass have the form of a function with saturation as well.

The dependences shown in Figs. 1–3 allow one to assume that the NDVI is not a direct characteristic of the structural properties of the vegetation cover of terrestrial ecosystems and is related to some other characteristics of the environment, for example, climatic ones.

Figure 4 shows the NDVI dependences on the total evaporation for (a) European Russia and (b) western Siberia. For European Russia, the total evaporation varies, on average, from 90 to 272 mm during the vegetation period, and, for western Siberia, this range is somewhat shifted toward smaller values and lies between 44 and 251 mm over the vegetation period. For the set of points obtained for European Russia, their density increases for NDVI values exceeding 0.5. A more pronounced dependence of the NDVI on the total evaporation over the vegetation period is characteristic of the set of points obtained for western Siberia. It is possible to assume that this result is associated with a more intense anthropogenic forcing on the ecosystems of European Russia than on those of western Siberia. The observed tendency for a decrease in NDVI values at a total evaporation exceeding 250 mm over the vegetation period (Fig. 4a) allows one to

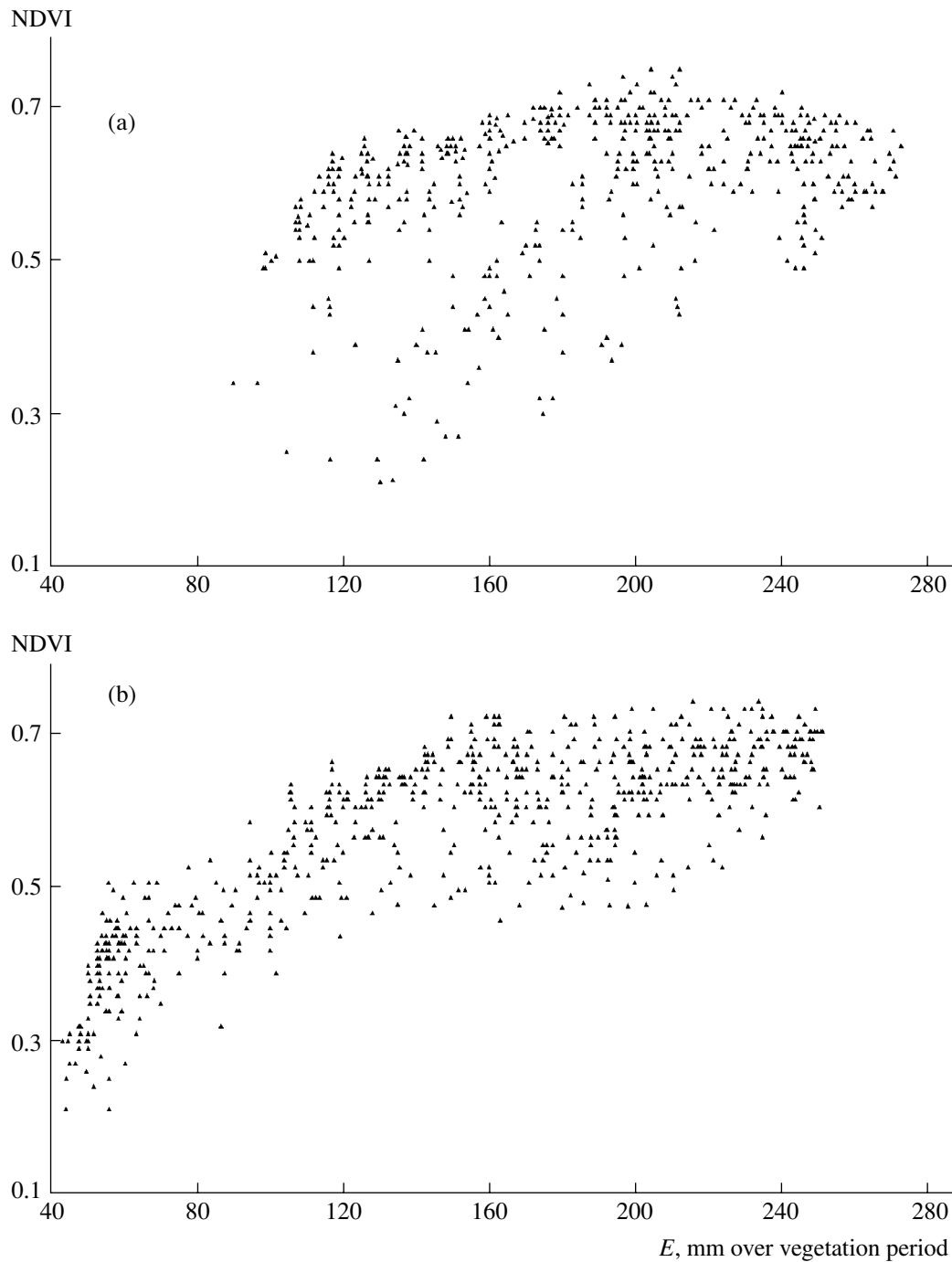


Fig. 4. Dependence of the NDVI on total evaporation (E) over the vegetation period for (a) European Russia and (b) western Siberia.

assume that the dependence under consideration has a parabolic form.

Figure 5 shows the NDVI dependences on the total incoming radiation over the vegetation period for (a) European Russia and (b) western Siberia. For European Russia, the total radiation varies from 1.33 to 2.06 GJ/m² over the vegetation period; for western Siberia, the range of these values is somewhat nar-

rower—from 1.44 to 1.98 GJ/m² over the vegetation period. For European Russia, the maximum density of points for the set under consideration is characteristic of total incoming radiation varying from 1.5 to 1.8 GJ/m² over the vegetation period, while, for western Siberia, a similar situation is characteristic of total incoming radiation varying from 1.56 to 1.72 GJ/m² over the vegetation period. It should be noted that, for both

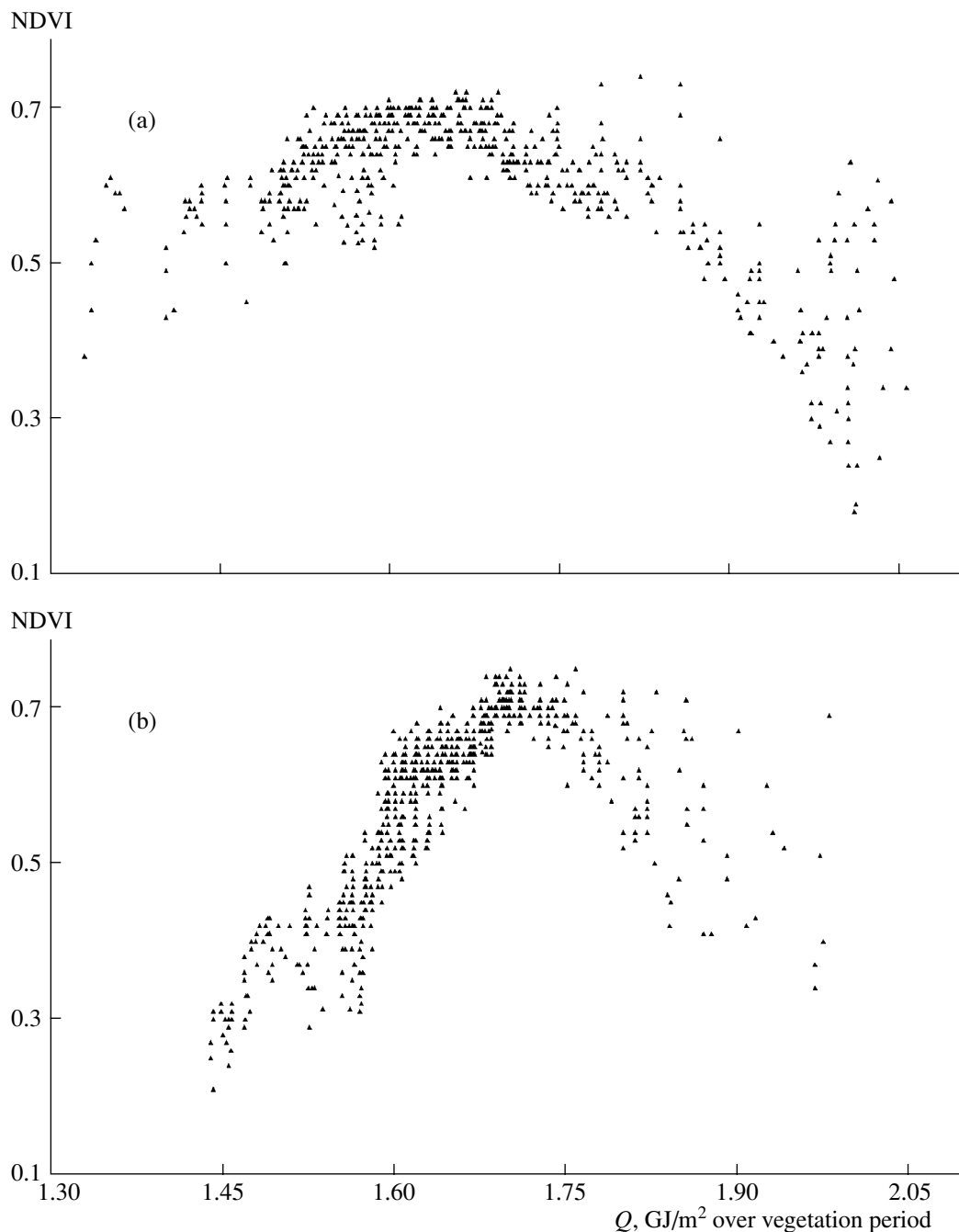


Fig. 5. Dependence of the NDVI on total incoming radiation (Q) over the vegetation period for (a) European Russia and (b) western Siberia.

European Russia and western Siberia, the NDVI dependence on the total radiation over the vegetation period has a parabolic form with peaks of about 1.64 and 1.7 GJ/m^2 , respectively. A gradual decrease in NDVI values from the parabola's peak is characteristic of European Russia, while a sufficiently abrupt decrease in NDVI values is characteristic of western Siberia.

Figure 6 presents the NDVI dependences on the mean (over the vegetation period) surface-air temperature for (a) European Russia and (b) western Siberia. The mean surface-air temperature over the period under consideration varies from 8.4 to 24.1°C for European Russia and from 4.8 to 21.6°C for western Siberia. For both European Russia and western Siberia, the dependence under consideration has a bell-

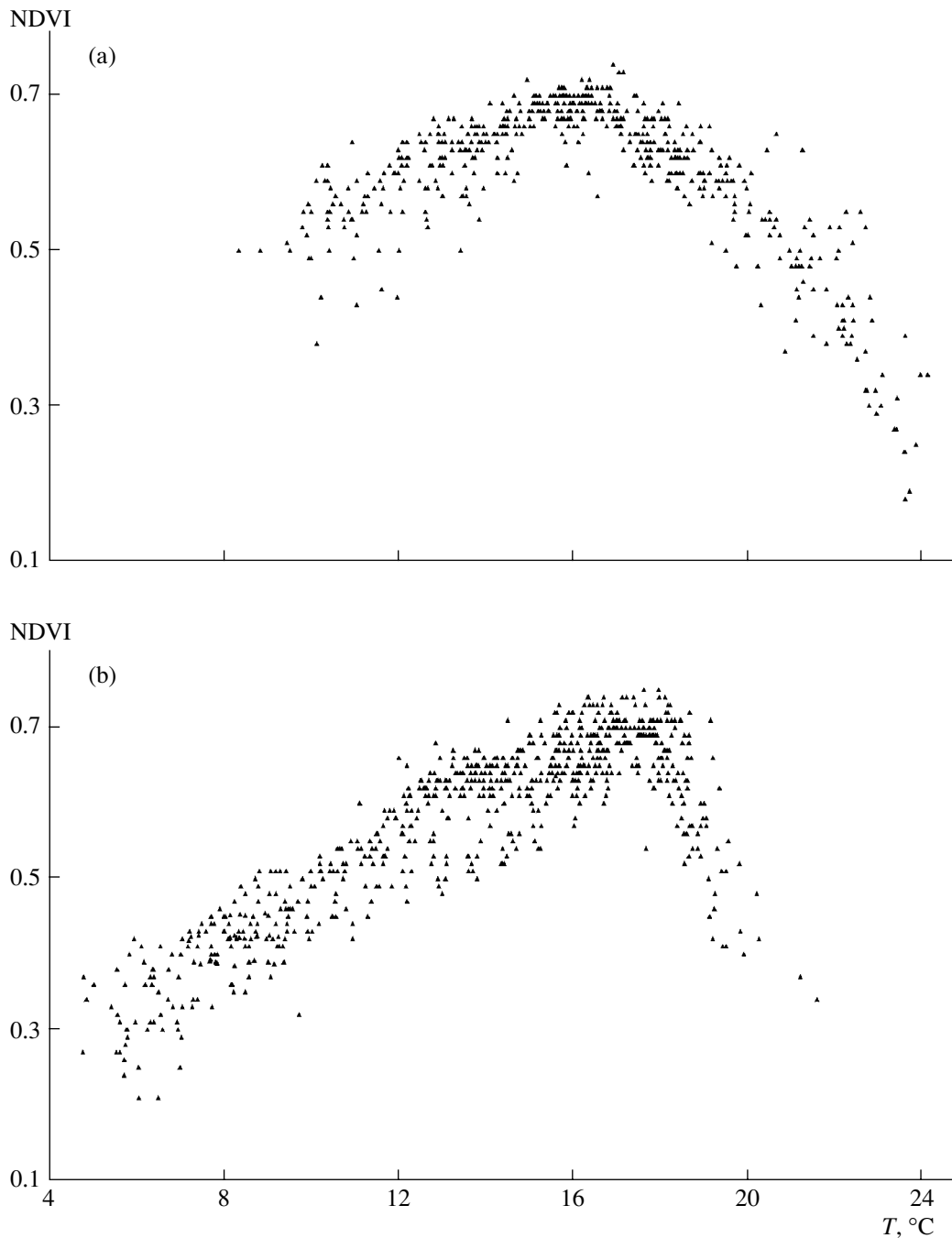


Fig. 6. Dependence of the NDVI on the mean (over the vegetation period) surface-air temperature (T) for (a) European Russia and (b) western Siberia.

shaped form with a peak at 16°C for European Russia and 17°C for western Siberia.

Table 2 presents the statistical characteristics of the NDVI for different ranges of variations in the climatic parameters under consideration. It follows from the sets of points given in Figs. 4–6 and the statistical characteristics given in Table 2 that the dependence of

NDVI on climatic parameters is more clearly defined than the dependence on the structural characteristics of vegetation cover. In particular, the dependence of NDVI on structural vegetation characteristics (especially at their small values) is highly variable. This finding reflects the fact that vegetation has a counter-intuitive behavior with regulatory properties, thus causing a high degree of unpredictability of the vege-

Table 2. Statistical characteristics of the NDVI for different ranges of variations in climatic parameters (symbols are the same as in Table 1)

Characteristic	Range of variations in total evaporation (mm over vegetation period)					
	0–80	80–120	120–160	160–200	200–240	240–280
European Russia						
$\bar{x}$	–	0.53	0.56	0.61	0.65	0.63
$x_{\min}$	–	0.24	0.21	0.30	0.43	0.49
$x_{\max}$	–	0.64	0.68	0.74	0.75	0.72
σ_x	–	0.09	0.12	0.10	0.63	0.05
v_x	–	15.9	20.6	15.9	9.7	8.2
Western Siberia						
$\bar{x}$	0.39	0.53	0.63	0.62	0.66	0.68
$x_{\min}$	0.21	0.32	0.47	0.46	0.48	0.53
$x_{\max}$	0.53	0.67	0.73	0.74	0.75	0.74
σ_x	0.07	0.07	0.06	0.07	0.05	0.4
v_x	16.8	13.4	8.8	10.5	8.2	5.6
Range of variations in total incoming radiation (GJ/m ² over vegetation period)						
	1.3–1.51	1.51–1.59	1.59–1.68	1.68–1.8	1.8–1.94	1.94–2.08
European Russia						
$\bar{x}$	0.56	0.63	0.67	0.63	0.55	0.41
$x_{\min}$	0.38	0.52	0.55	0.56	0.40	0.18
$x_{\max}$	0.66	0.70	0.72	0.73	0.74	0.63
σ_x	0.06	0.05	0.03	0.04	0.08	0.11
v_x	9.8	7.3	4.6	6.0	14.1	27.7
Western Siberia						
$\bar{x}$	0.35	0.44	0.62	0.69	0.58	0.47
$x_{\min}$	0.21	0.29	0.47	0.58	0.41	0.34
$x_{\max}$	0.43	0.63	0.74	0.75	0.72	0.69
σ_x	0.06	0.06	0.05	0.03	0.08	0.12
v_x	16.5	13.4	8.4	5.0	14.5	25.1
Range of variations in mean (over vegetation period) surface-air temperature (°C)						
	4–8	8–12	12–15	15–18	18–22	22–25
European Russia						
$\bar{x}$	–	0.55	0.63	0.67	0.57	0.37
$x_{\min}$	–	0.38	0.50	0.57	0.37	0.18
$x_{\max}$	–	0.64	0.72	0.74	0.69	0.55
σ_x	–	0.05	0.04	0.03	0.07	0.09
v_x	–	9.1	6.2	4.3	11.8	25.3
Western Siberia						
$\bar{x}$	0.36	0.47	0.61	0.67	0.60	–
$x_{\min}$	0.21	0.32	0.47	0.54	0.34	–
$x_{\max}$	0.46	0.60	0.71	0.75	0.74	–
σ_x	0.06	0.06	0.05	0.04	0.09	–
v_x	16.8	12.2	8.1	6.1	15.7	–

tation's response to environmental changes of both natural and anthropogenic character.

CONCLUSIONS

Variations in NDVI values in relation to the climatic parameters and structural characteristics of vegetation cover were analyzed on the basis of data obtained for European Russia and western Siberia.

It has been shown that the relationship between the NDVI and the structural vegetation characteristics, such as the overground net primary production and the overground phytomass, is described by a curve that flattens out rapidly. The relationships under consideration have a wide scatter in NDVI values at small values of net primary production and phytomass when vegetation canopy is not closed and, therefore, reflection from soil is strong. With an increase in the growth rate of green-vegetation mass, with an increase in the overground phytomass, and with complications of the adaptive and edification structures of vegetation cover, the scatter in NDVI values decreases. Comparison of the latitudinal distributions of NDVI values, overground organic production, and overground phytomass has shown that, in these distributions, the maxima of the indicated quantities and the rates of their variations do not coincide. This finding suggests that the NDVI weakly reflects the structural characteristics of vegetation cover.

The dependences of the NDVI on climatic parameters, such as the total incoming radiation, mean surface-air temperature, and total evaporation over the vegetation period, have a bell-shaped form. It follows from the relationship obtained between the NDVI and the total incoming radiation over the vegetation period that the NDVI maximum lies within the range 1.5–1.7 GJ/m² for European Russia and within the range 1.65–1.72 GJ/m² for western Siberia. The dependence of NDVI on the mean surface-air temperature over the vegetation period is characterized by an NDVI maximum approximately within the range from 16 to 17°C for both European Russia and western Siberia. For the areas under consideration, the maximum values of the NDVI treated as a function of the total evaporation over the vegetation period are reached at total evaporation values of 160 to 230 mm. It should be noted that, for European Russia, the dependences of NDVI on the total incoming radiation and mean surface-air temperature have bell-shaped forms with significantly flatter peaks than similar dependences for western Siberia.

The above analysis has shown that the NDVI is not an indicator for the values of overground NPP and overground phytomass within the sufficiently large areas under consideration. Thus, the use of the NDVI in models for determining the structural characteristics of the vegetation cover of terrestrial ecosystems in

European Russia and western Siberia is rather problematic. At the same time, the use of the NDVI during prediction of climatic parameters such as the surface-air temperature and the total incoming radiation may be successful.

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